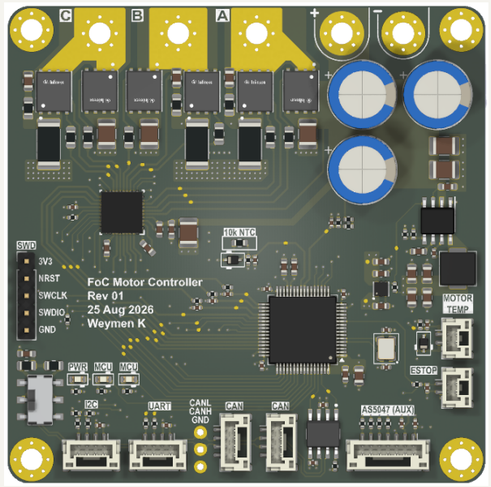
FoC Motor Controller

Variant: Prototype

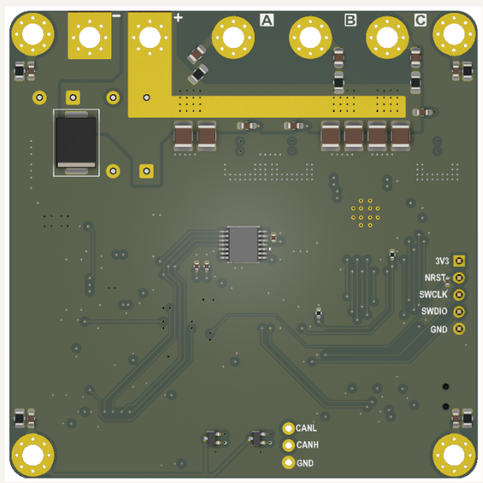
28th August 2026
Rev 01

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6	Motor Control - Driver	16	.....	26	.....	36	.....
7	MOSFETs	17	.....	27	.....	37	.....
8	Sensing - Position	18	.....	28	.....	38	.....
9	Sensing - Battery	19	.....	29	.....	39	.....
10	Sensing - Temperature	20	.....	30	.....	40	.....

TOP VIEW



BOTTOM VIEW



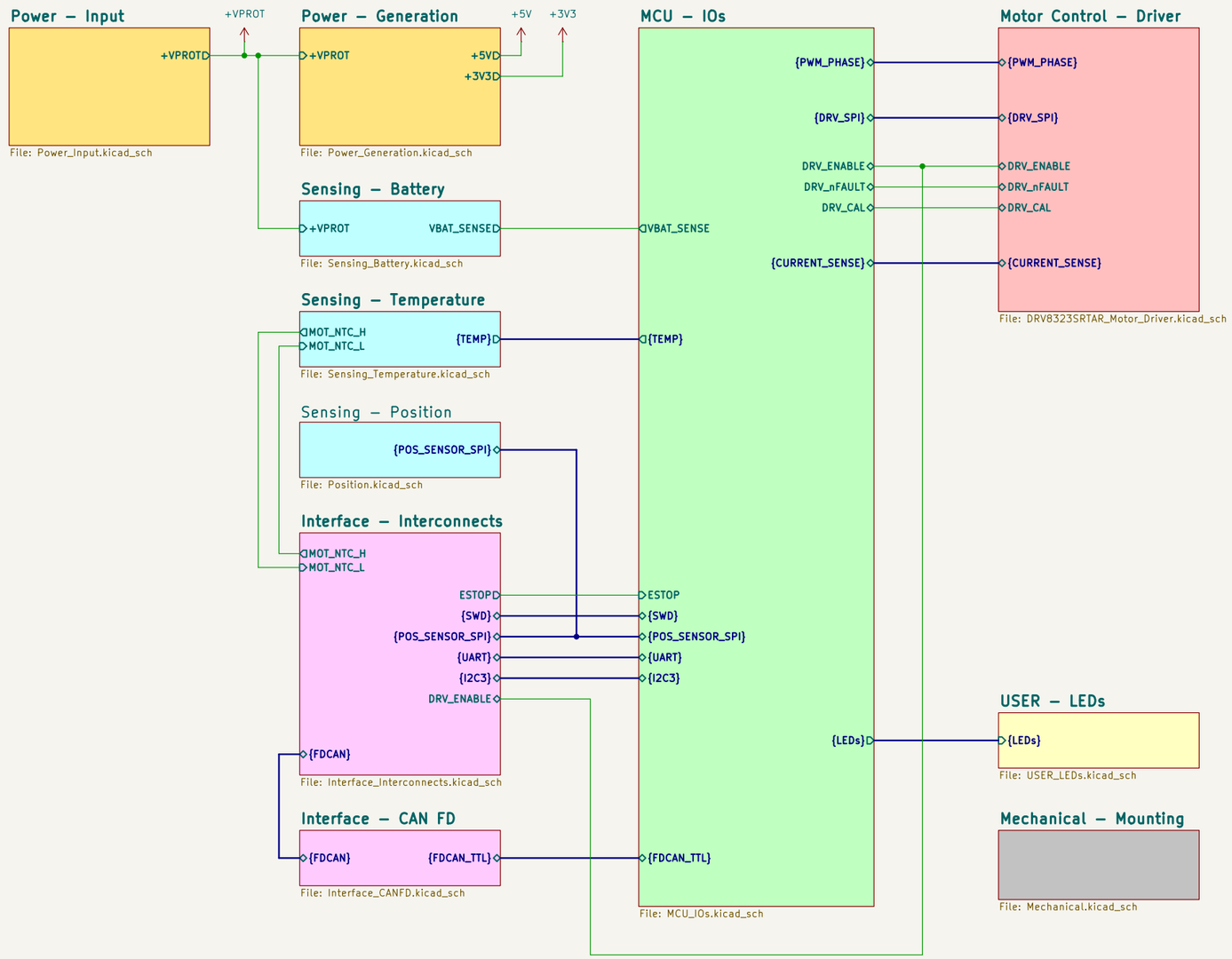
NOTES

Not fitted components are marked as ✗

References:
Vincent Nguyen: https://github.com/nguyen-v/amulet_controller_kibot
SIROJU: https://www.youtube.com/@srj4555

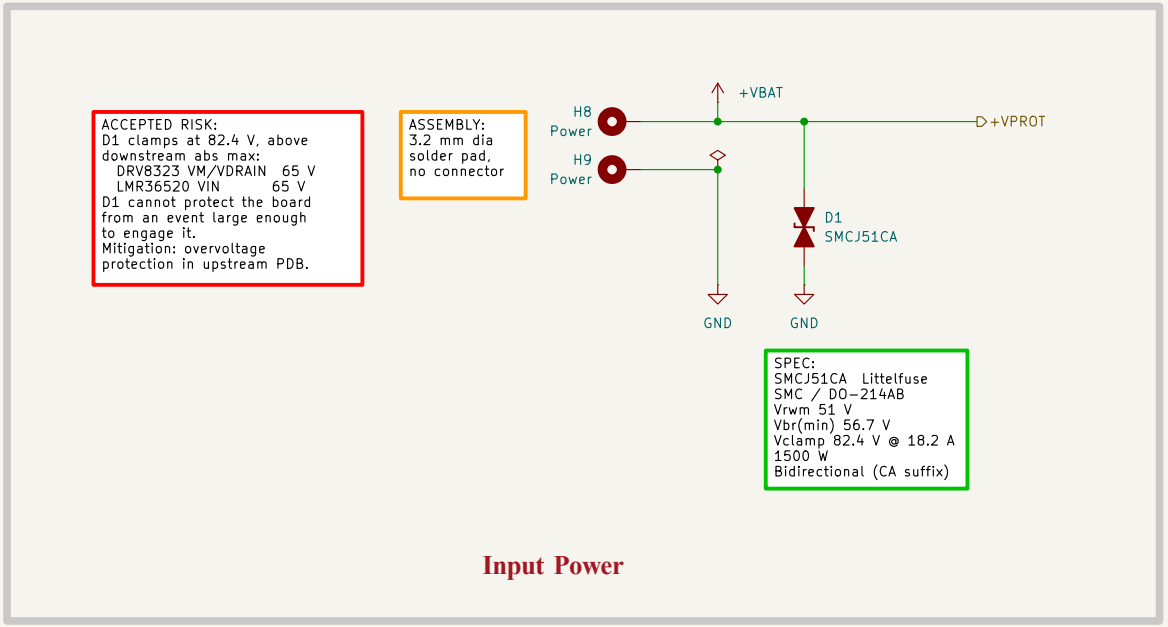
Comments:	Company: -			Variant:
	Board Name: **FoC Motor Controller**			Project Name: **FoC Motor Controller**
Sheet Title: Cover Page	File Name: FoC Motor Controller.kicad_sch	Designer: Weymen Koo	Date: 28th August 2026	Revision: 01
Sheet Path: /		Reviewer: N/A	Size: **A3**	Sheet: **1** of **14**

[2] Project Architecture



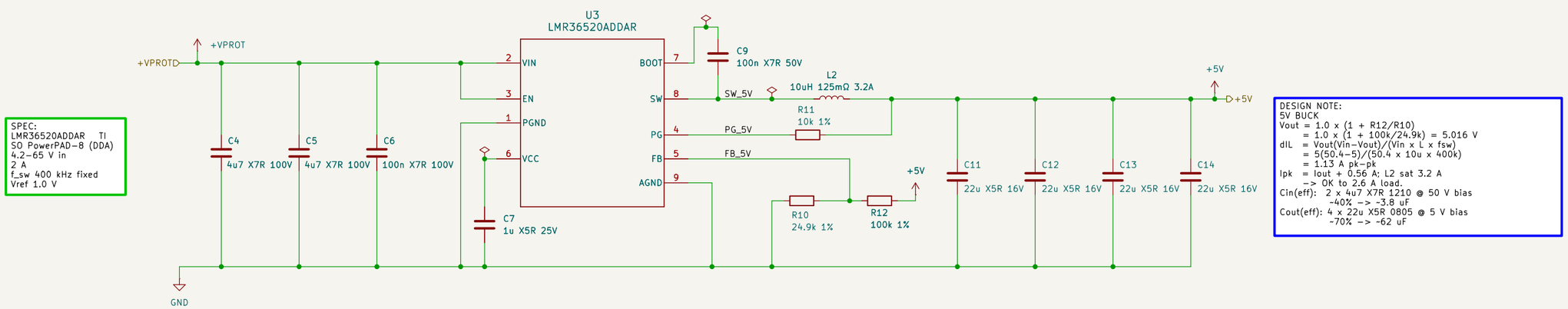
Comments:	Company: -		Variant:	
	Board Name: **FoC Motor Controller**		Project Name: **FoC Motor Controller**	
Sheet Title: Project Architecture	File Name: Project Architecture.kicad_sch	Designer: Weymen Koo	Date: 28th August 2026	Revision: 01
Sheet Path: /Project Architecture/		Reviewer: N/A	Size: **A3**	Sheet: **2** of **14**

[3] Power - Input



	Comments:	Company: -		Variant:
		Board Name: **FoC Motor Controller**		Project Name: **FoC Motor Controller**
	Sheet Title: Power - Input	File Name: Power_Input.kicad_sch	Designer: Weymen Koo	Date: 28th August 2026 / Revision: 01
	Sheet Path: /Project Architecture/Power - Input/		Reviewer: N/A	Size: **A4** / Sheet: **3** of **14**

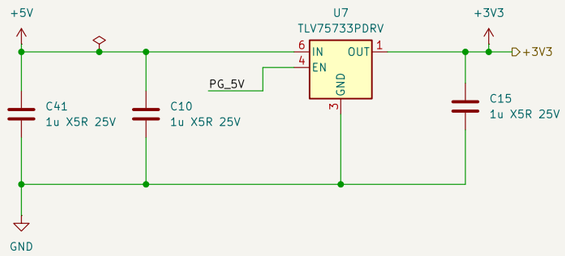
[4] Power - Generation



SPEC:
LMR36520ADDAR TI
SO PowerPAD-8 (DDA)
4.2–65 V in
2 A
f_sw 400 kHz fixed
Vref 1.0 V

DESIGN NOTE:
5V BUCK
Vout = 1.0 x (1 + R12/R10)
= 1.0 x (1 + 100k/24.9k) = 5.016 V
dIL = Vout(Vin−Vout)/(Vin x L x fsw)
= 5(50.4−5)/(50.4 x 10u x 400k)
= 1.13 A pk-pk
Ipk = Iout + 0.56 A; L2 sat 3.2 A
-> OK to 2.6 A load.
Cin(eff): 2 x 4u7 X7R 1210 @ 50 V bias
~40% -> ~3.8 uF
Cout(eff): 4 x 22u X5R 0805 @ 5 V bias
~70% -> ~62 uF

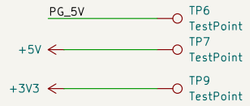
5V Buck Converter



DESIGN NOTE:
3V3 BUDGET AND LDO THERMAL
Pd = (5.02 − 3.30) x 500 mA = 0.86 W
Theta-JA (WSON-6, 2 oz, pad vias) ~= 78 C/W
dT = 67 C -> TJ = 92 C at 25 C ambient
(limit 125 C). 500 mA is the budget ceiling;
keep the real 3V3 load well below it in a
warm ambient.

SPEC:
TLV75733PDRV TI
WSON-6 2x2 (DRV)
Vin 2.5–5.5 V
1 A
Vdo ~240 mV @ 1 A
fixed 3.3 V
EN gated on PG_5V:
3V3 cannot precede 5V.

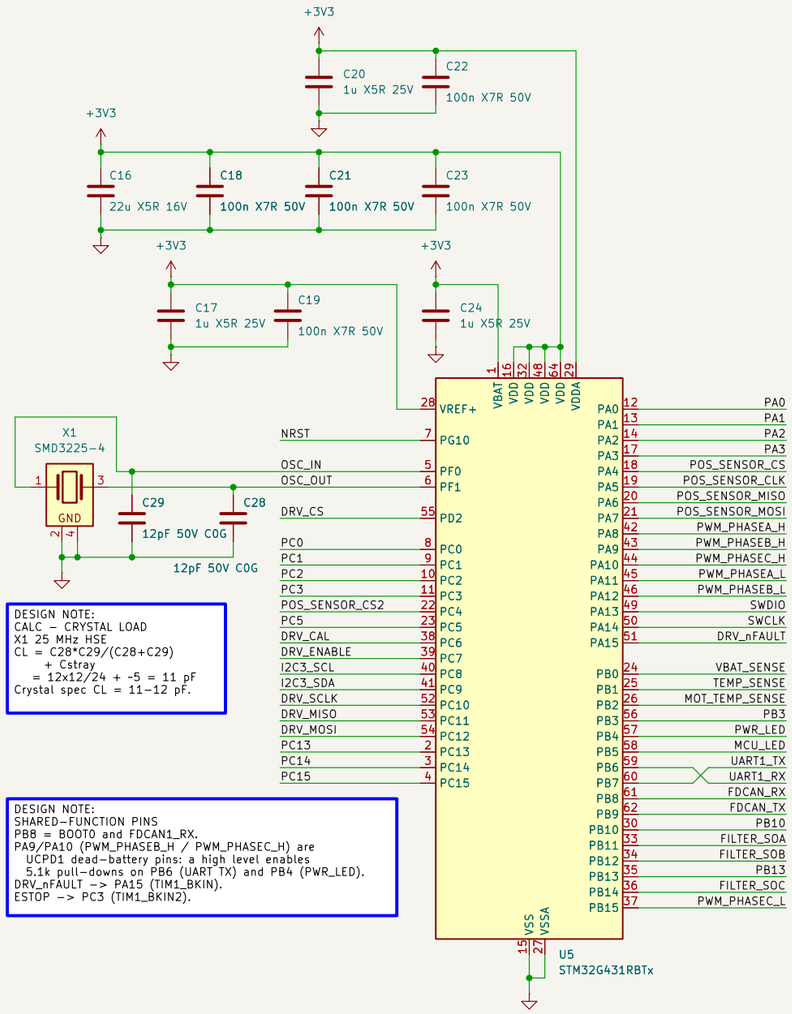
3V3 Linear Regulator



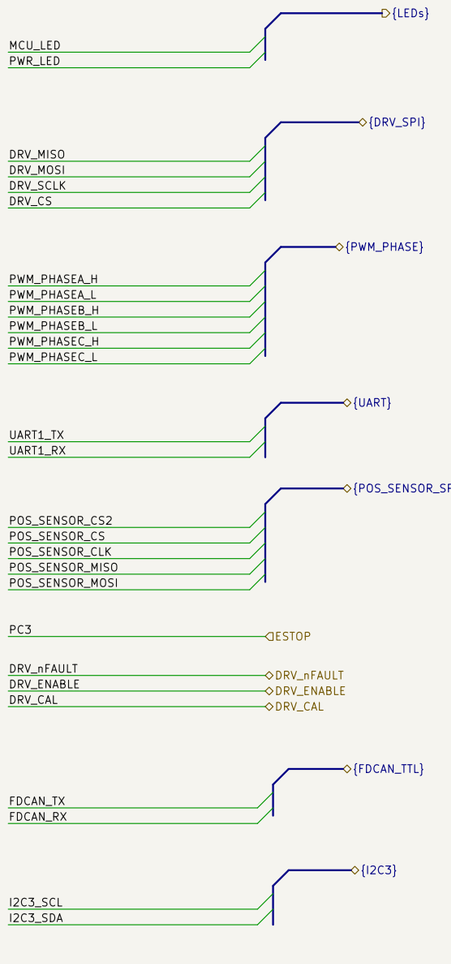
Testpoints

Comments:	Company: -		Variant:	
	Board Name: **FoC Motor Controller**		Project Name: **FoC Motor Controller**	
Sheet Title: Power - Generation	File Name: Power_Generation.kicad_sch	Designer: Weymen Koo	Date: 28th August 2026	Revision: 01
Sheet Path: /Project Architecture/Power - Generation/		Reviewer: N/A	Size: **A3**	Sheet: **4** of **14**

[5] MCU - IOs

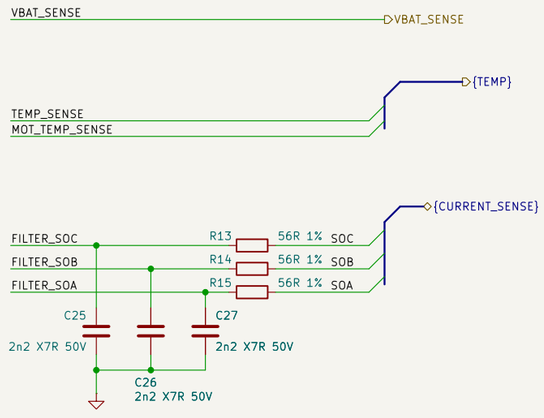


MCU - STM32G431RBT6



Digital IO

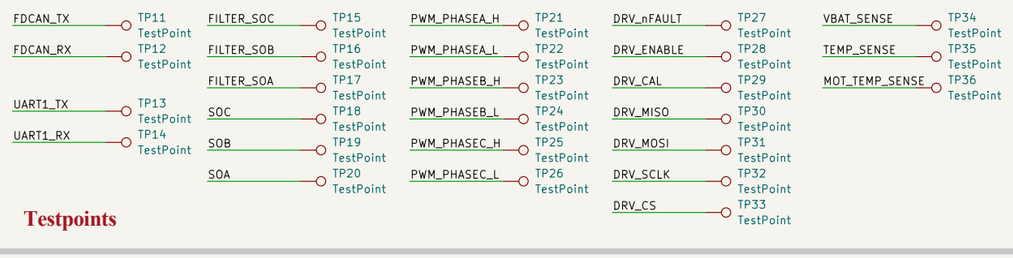
SPEC:
ADC CHANNEL MAP
FILTER_SOA PB11 ADC1_IN14 / ADC2_IN14 (both)
FILTER_SOB PB12 ADC1_IN11 (ADC1 only)
FILTER_SOC PB14 ADC1_IN5 (ADC1 only)
VBAT_SENSE PB0 ADC1_IN15
TEMP_SENSE PB1 ADC1_IN12
MOT_TEMP PB2 ADC2_IN12
SOB and SOC are both ADC1-only: phases B and C
cannot be sampled simultaneously.



Analog IO



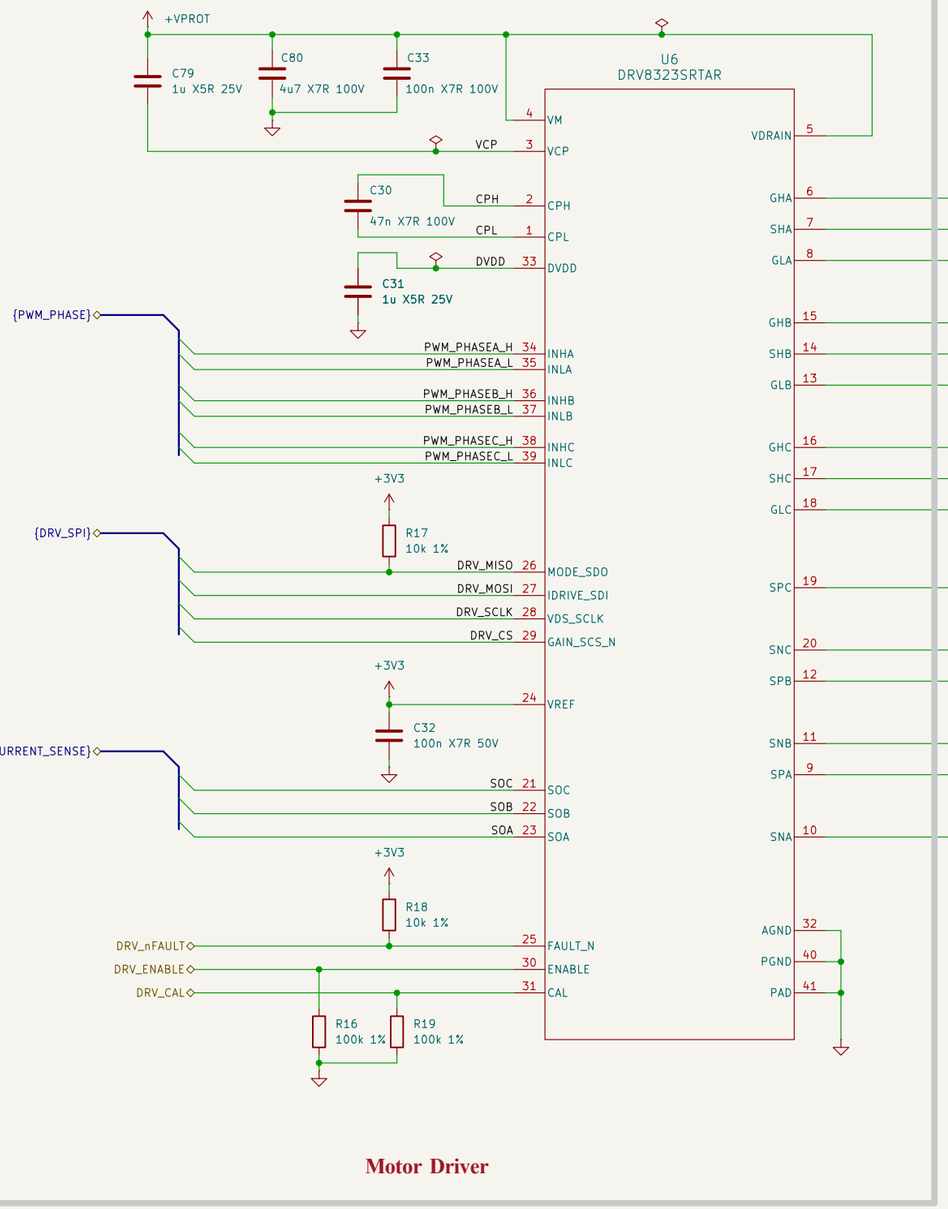
SWD



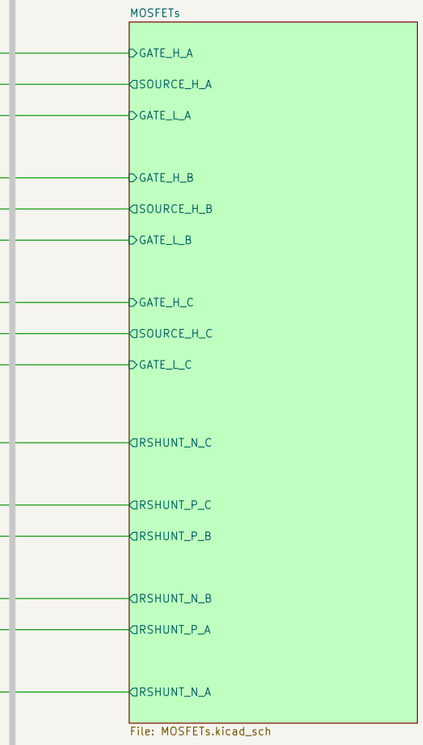
Testpoints

Comments:	Company: -		Variant:
	Board Name: **FoC Motor Controller**		Project Name: **FoC Motor Controller**
Sheet Title: MCU - IOs	File Name: MCU_IOs.kicad_sch	Designer: Weymen Koo	Date: 28th August 2026 / Revision: 01
Sheet Path: /Project Architecture/MCU - IOs/		Reviewer: N/A	Size: A3 / Sheet: 5 of 14

[6] Motor Driver - DRV8323SRTAR



Motor Driver



Inverter

SPEC:
DRV8323SRTAR TI WQFN-40 6x6 (RTA)
VM 6-60 V rec, 65 V abs max
3x CSA
SPI
1 A/2 A gate drive
Digital pin abs max 5.75 V (independent of VM)
SPx/SNx: +/-1 V continuous, +/-3 V transient 200 ns

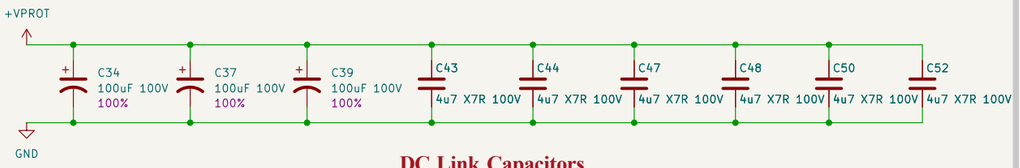
DESIGN NOTE:
CALC - CHARGE PUMP (high-side gates)
I_VCP = 3 x Qg x f_sw = 3 x 55 nC x 40 kHz = 6.6 mA
DRV8323 VCP capacity 25 mA -> 3.8x margin
Max FET Qg at 2x margin -= 104 nC

REV 1 NOTE:
KNOWN ISSUES (no Rev 1 change)
SPC/SNC swapped vs phases A/B (SPC -> RSHUNT_N_C):
phase C CSA output is sign-inverted.
SDO is open-drain, R17 10k pull-up limits SPI clock.
DRV_CS has no pull-up.

Test Points & Design Notes

Comments:	Company: -		Variant:	
	Board Name: **FoC Motor Controller**		Project Name: **FoC Motor Controller**	
Sheet Title: Motor Control - Driver	File Name: DRV8323SRTAR_Motor_Driver.kicad_sch	Designer: Weymen Koo	Date: 28th August 2026	Revision: 01
Sheet Path: /Project Architecture/Motor Control - Driver/		Reviewer: N/A	Size: **A3**	Sheet: **6** of **14**

[7] MOSFETs

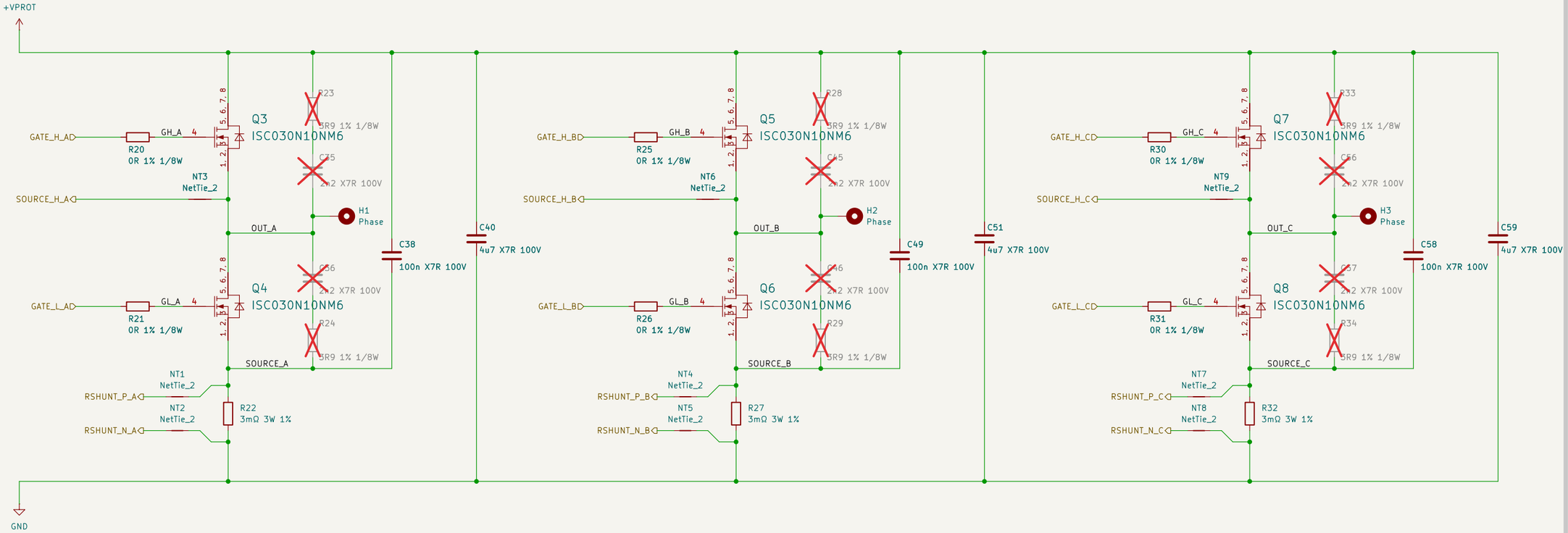


DC Link Capacitors

SPEC:
Q3–Q8 ISC030N10NM6 Infineon OptiMOS 6
PG–TDSON–8 (5 x 6 mm)
100 V, 3.0 mR max @ VGS 10 V
Qg 55 nC, Qgd 14 nC, tr/tf 4.5/5.3 ns
R22/R27/R32 3 mR 3 W 1% low–side shunt
C34/C37/C39 100 uF 100 V polymer

DESIGN NOTE:
CALC – DC LINK
Bulk 3 x 100 uF polymer = 300 uF
MLCC 9 x 4u7 X7R 100 V (6 bank + C40/C51/C59)
-40% left at 50 V bias –> ~17 uF effective;
sets commutation–loop ESL, not ripple capacity.
Ripple ~7.8 A RMS needed @ 10 A/phase
vs 3 x 2.5 A = 7.5 A rated (~96%).
Sizing called for 5 pcs; 3 fitted on Rev 1.
Measure per–cap ripple before production.

DESIGN NOTE:
GATE, SNUBBER AND SHUNT
Gate R = 0R: slew set by DRV8323 IDRIVE.
R23/C35 etc. (3R9 + 2n2) D–S RC snubbers, DNP:
fit only if ringing is observed.
Shunts Kelvin–sensed via NT1/2, NT4/5, NT7/8.
NT3/6/9 = SHx high–side source sense.
CSA range = 1.65 V / (G x 3 mR)
= +/–27.5 A at G = 20 V/V



Three-Phase Inverter Bridge

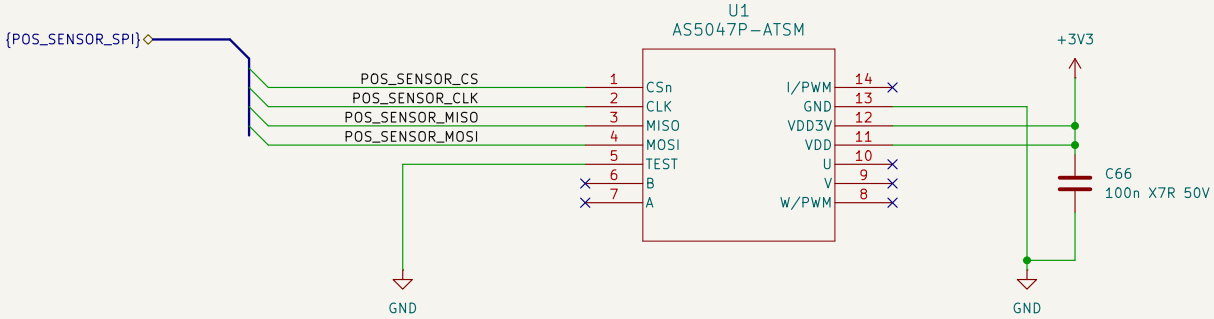
Comments:	Company: -			Variant:
	Board Name: **FoC Motor Controller**			Project Name: **FoC Motor Controller**
Sheet Title: MOSFETs	File Name: MOSFETs.kicad_sch	Designer: Weymen Koo	Date: 28th August 2026	Revision: 01
Sheet Path: /Project Architecture/Motor Control - Driver/MOSFETs/		Reviewer: N/A	Size: **A3**	Sheet: **7** of **14**

[8] Sensing - Position

SPEC:
U1 AS5047P-ATSM ams TSSOP-14
14-bit (16384 counts/rev) | up to 28 krpm | DAEC
SPI slave, mode 1 (CPOL=0, CPHA=1), up to 10 MHz, MSB first

ASSEMBLY:
MAGNET PLACEMENT (datasheet reference values)
Diametric magnet, ref. N35H 8 mm dia x 3 mm.
Air gap ~2.0 mm, Bz 35-70 mT.
Centre on U1 package within 0.5 mm
(INL +/-1.2 deg at 500 um offset).
Keep ferrous hardware and phase/bus copper away.

DESIGN NOTE:
VDD and VDD3V tied for 3.3 V operation
TEST (pin 5) tied to GND - required
A/B/I/U/V/W left open - permitted (datasheet pin note 5)

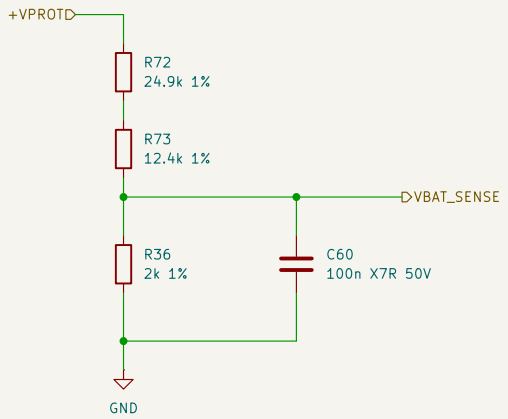


Absolute Position Sensor

Comments:	Company: -		Variant:	
	Board Name: **FoC Motor Controller**		Project Name: **FoC Motor Controller**	
Sheet Title: Sensing - Position	File Name: Position.kicad_sch	Designer: Weymen Koo	Date: 28th August 2026	Revision: 01
Sheet Path: /Project Architecture/Sensing - Position/		Reviewer: N/A	Size: **A4**	Sheet: **8** of **14**

[9] Sensing - Battery

DESIGN NOTE:
  Ratio = R36/(R72+R73+R36)
        = 2/(24.9+12.4+2) = 0.05089
    at 50.4 V -> 2.565 V
    at 56.0 V -> 2.850 V
    clips at VREF+ = 3.3 V when Vbus = 64.8 V
  Divider current at 50.4 V = 1.283 mA
    (continuous pack drain)
  P(R72) = 41 mW, P(R73) = 20 mW
  Rth = (R72+R73) || R36 = 1.90 k
  tau = 190 us, fc = 838 Hz

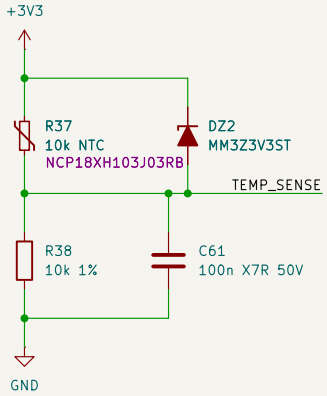


Battery Voltage Sensor

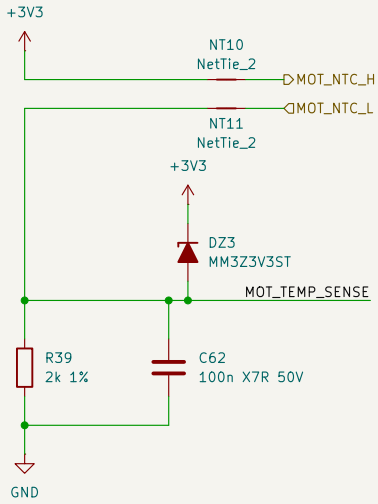
Comments:	Company: -		Variant:	
	Board Name: **FoC Motor Controller**		Project Name: **FoC Motor Controller**	
Sheet Title: Sensing - Battery	File Name: Sensing_Battery.kicad_sch	Designer: Weymen Koo	Date: 28th August 2026	Revision: 01
Sheet Path: /Project Architecture/Sensing - Battery/		Reviewer: N/A	Size: **A4**	Sheet: **9** of **14**

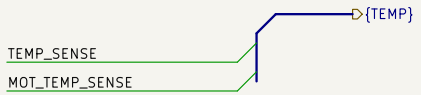
[10] Sensing - Temperature

DESIGN NOTE:
CALC – 10K NTC ON-BOARD (R37)
NCP18XH103J03RB, R38 10k to GND
V = 3.3 x R38/(NTC + R38)
−10 C (~42k) –> 0.63 V
25 C (10k) –> 1.65 V
100 C (~680R) –> 3.09 V
Self-heat: I = 165 uA, P = 0.27 mW,
<0.1 C rise. Negligible.



On-Board 10k NTC



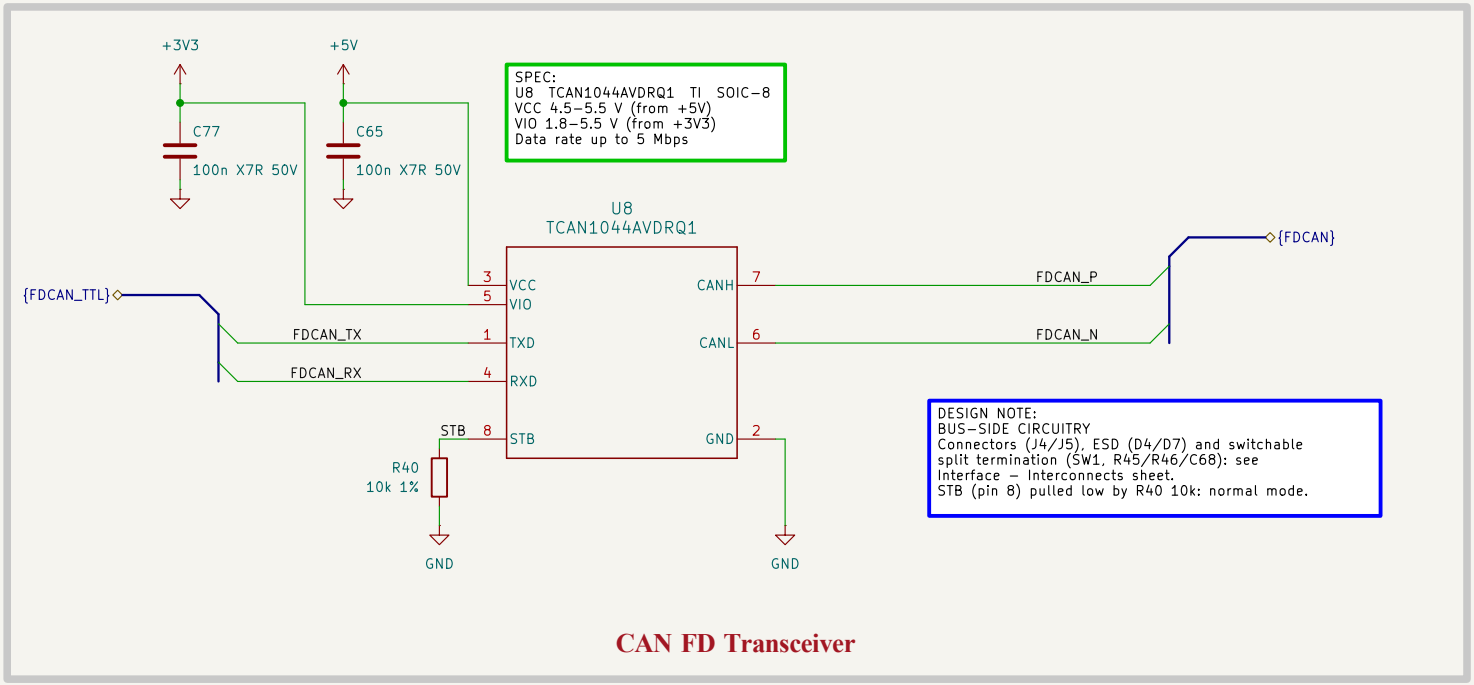


DESIGN NOTE:
CALC – MOTOR NTC (external NTC on top,
R39 2k to GND)
V = 3.3 x 2k/(NTC + 2k)
25 C (10k) –> 0.55 V
100 C (~680R) –> 2.46 V
DZ3 clamps the off-board line to 3.3 V.

Motor NTC (via J10)

Comments:	Company: -		Variant:	
	Board Name: **FoC Motor Controller**		Project Name: **FoC Motor Controller**	
Sheet Title: Sensing - Temperature	File Name: Sensing_Temperature.kicad_sch	Designer: Weymen Koo	Date: 28th August 2026	Revision: 01
Sheet Path: /Project Architecture/Sensing - Temperature/		Reviewer: N/A	Size: **A4**	Sheet: **10** of **14**

[11] Interface - CAN FD



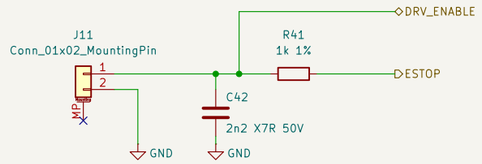
CAN FD Transceiver

Comments:	Company: -		Variant:	
	Board Name: **FoC Motor Controller**		Project Name: **FoC Motor Controller**	
Sheet Title: Interface - CAN FD	File Name: Interface_CANFD.kicad_sch	Designer: Weymen Koo	Date: 28th August 2026	Revision: 01
Sheet Path: /Project Architecture/Interface - CAN FD/		Reviewer: N/A	Size: **A4**	Sheet: **11** of **14**

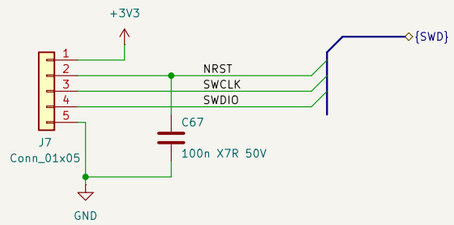
[12] Interface - Interconnects

DESIGN NOTE:
ESD PROTECTION
CAN bus: D4/D7 PESD2CANFD24V on CANH/CANL.
SWD, UART, I2C, AUX2, NTC and E-stop connectors unprotected (prototype, lab use).

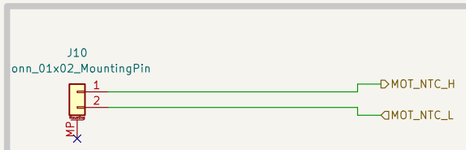
REV 1 NOTE:
E-STOP NOT FUNCTIONAL AS DRAWN
J11-1 connects directly to DRV_ENABLE; R41/C42 feed the same node into PC3 (ESTOP).
- Unplugged: R16 100k holds DRV_ENABLE low (driver off = fail-safe).
- J11 looped to GND: shorts the MCU DRV_ENABLE output; driver cannot be enabled.
- PC3 only mirrors DRV_ENABLE.
Leave J11 unplugged; bench supply switch is the E-stop. Rev 2: separate E-stop input.



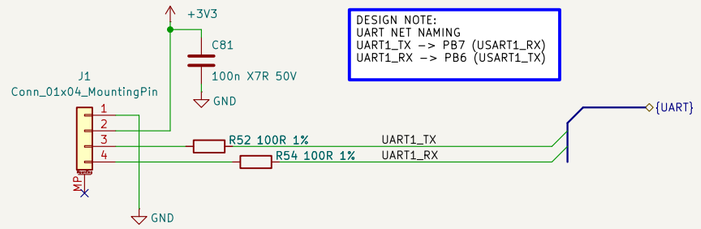
E-Stop

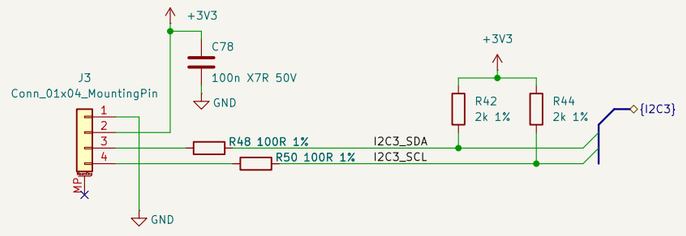


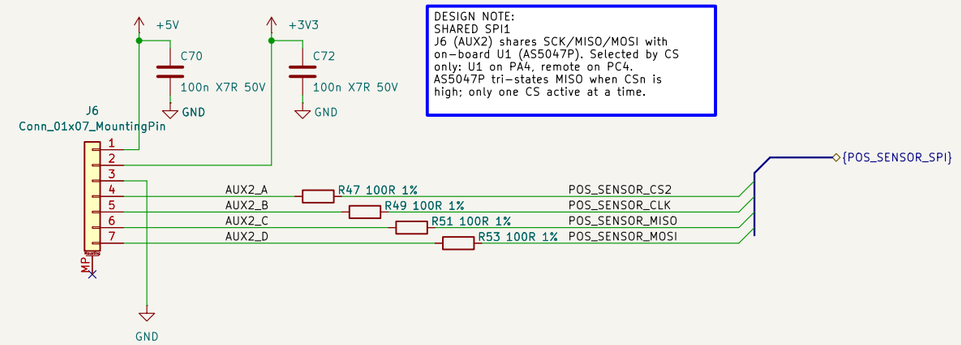
SWD (2.54mm Pin Header)



Temperature

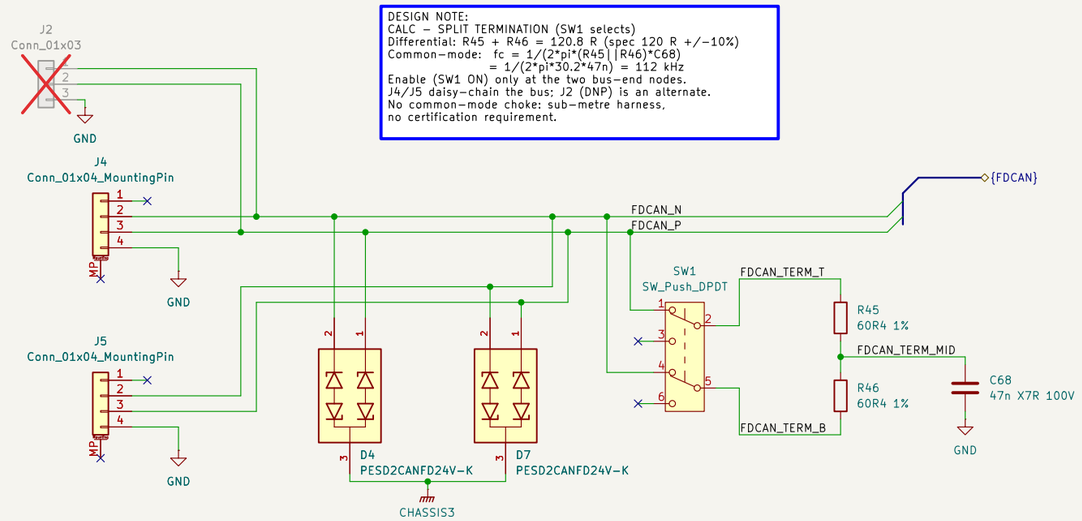






AUX2 Connector (Position Sensor SPI)

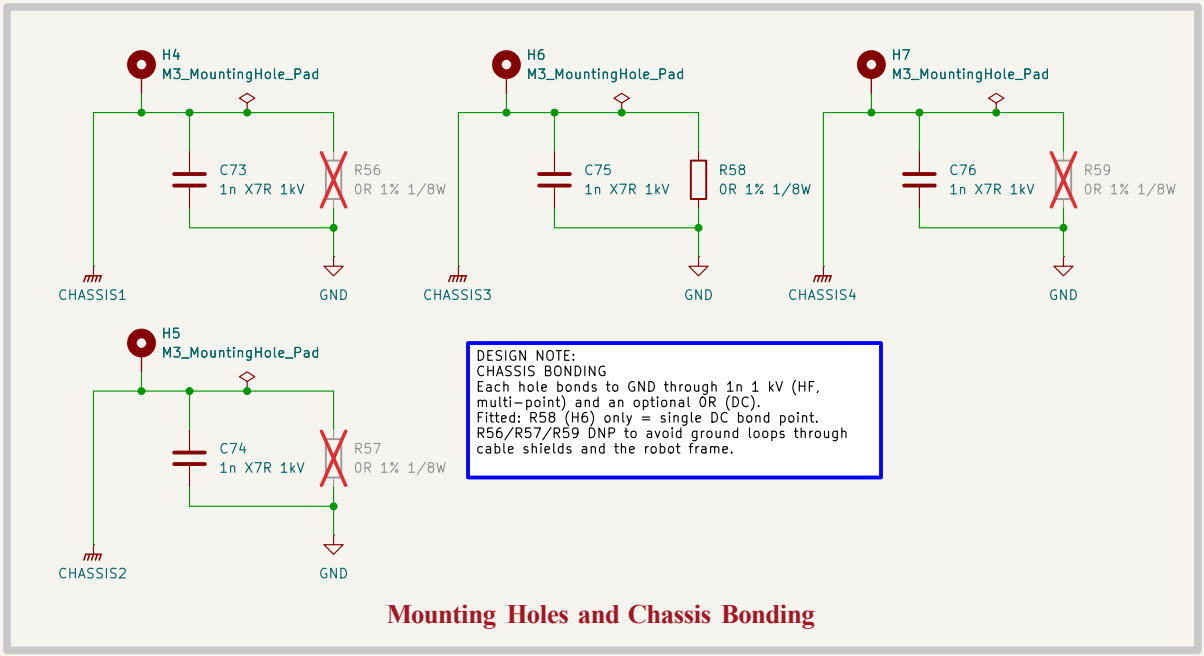
DESIGN NOTE:
CALC - SPLIT TERMINATION (SW1 selects)
Differential: R45 + R46 = 120.8 R (spec 120 R +/-10%)
Common-mode: fc = 1/(2*pi*(R45||R46)*C68)
= 1/(2*pi*30.2*47n) = 112 kHz
Enable (SW1 ON) only at the two bus-end nodes.
J4/J5 daisy-chain the bus; J2 (DNP) is an alternate.
No common-mode choke: sub-metre harness, no certification requirement.



FD_CAN Connectors and Termination

Comments:	Company: -		Variant:	
	Board Name: **FoC Motor Controller**		Project Name: **FoC Motor Controller**	
Sheet Title: Interface - Interconnects	File Name: Interface_Interconnects.kicad_sch	Designer: Weymen Koo	Date: 28th August 2026	Revision: 01
Sheet Path: /Project Architecture/Interface - Interconnects/		Reviewer: N/A	Size: **A3**	Sheet: **12** of **14**

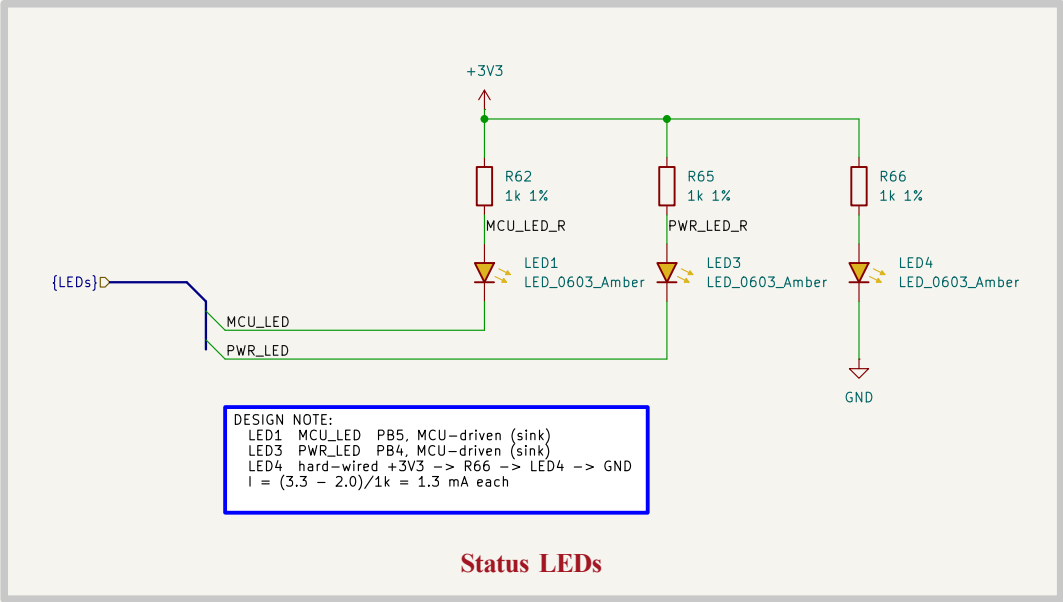
[13] Mechanical - Mounting Holes



Mounting Holes and Chassis Bonding

Comments:	Company: -		Variant:	
	Board Name: **FoC Motor Controller**		Project Name: **FoC Motor Controller**	
Sheet Title: Mechanical - Mounting	File Name: Mechanical.kicad_sch	Designer: Weymen Koo	Date: 28th August 2026	Revision: 01
Sheet Path: /Project Architecture/Mechanical - Mounting/		Reviewer: N/A	Size: **A4**	Sheet: **13** of **14**

[14] User - LEDs



	Comments:	Company: -		Variant:
		Board Name: **FoC Motor Controller**		Project Name: **FoC Motor Controller**
	Sheet Title: USER - LEDs	File Name: USER_LEDs.kicad_sch	Designer: Weymen Koo	Date: 28th August 2026 / Revision: 01
	Sheet Path: /Project Architecture/USER - LEDs/		Reviewer: N/A	Size: **A4** / Sheet: **14** of **14**